

Strict cumulative universes in $(\infty, 1)$ -toposes

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Abstract

We show that type theory with a cumulative hierarchy of univalent universes has interpretation in all Grothendieck $(\infty, 1)$ -toposes. This is achieved by adapting the arguments for the Grothendieck 1-toposic case to the construction for $(\infty, 1)$ -toposes. A crucial lemma is *realignment*, which allows to choose suitable codes for type connectives given appropriate universe inclusions. Here, we exhibit such universe inclusions by a careful analysis of the small object construction.

1 Preliminaries

1.1 Universes in model categories

Let $\mathcal{E}$ be a model category. Following Voevodsky, a universe in $\mathcal{E}$ should be a fibration $\pi : \tilde{\mathbf{U}} \rightarrow \mathbf{U}$ together with chosen pullbacks along any code $A : \Gamma \rightarrow U$, which models the extension of Γ with the decoded type of A . Those chosen pullbacks are called display maps and should correspond to the fibrations of $\mathcal{E}$. Hence, our universe $\mathbf{U}$ should be a sort of "fibration classifier", but this leads to two issues: size and coherence. The latter comes down to the fact that our fibrations form a pseudofunctorial class of chosen pullbacks instead of a strict one. Namely, we have the following pseudofunctors.

Definition 1.1 (Core of self-indexing). The core of self-indexing of $\mathcal{E}$ is the following groupoid-valued pseudofunctor.

$$\mathcal{E}^{\text{op}} \xrightarrow{\mathbb{E}} \mathcal{GPD}$$

$$\begin{array}{ccc} B & \longmapsto & (\mathcal{E}/_B)^\cong \\ f \uparrow & & \downarrow f^* \\ A & \longmapsto & (\mathcal{E}/_A)^\cong \end{array}$$

If $\mathcal{E}$ has a pullback-stable class of fibrations $\mathcal{F}$, then we also have:

$$\mathcal{E}^{\text{op}} \xrightarrow{\mathbb{F}} \mathcal{GPD}$$

$$\begin{array}{ccc} B & \longmapsto & (\mathcal{F}/_B)^\cong \\ f \uparrow & & \downarrow f^* \\ A & \longmapsto & (\mathcal{F}/_A)^\cong \end{array}$$

obtained by restricting $\mathbb{E}$ to the full subgroupoid of fibrations over B .

It might be tempting to try to find a *strong* classifier for fibrations, and ask the universe $\pi : \tilde{\mathbf{U}} \rightarrow \mathbf{U}$ to define a pseudonatural isomorphism $\mathcal{E}(B, \mathbf{U}) \cong \mathbb{F}(B)$, but this would be too restrictive, as this would be an isomorphism between a set and a groupoid. Therefore, our universe should be a *weak* classifier for fibrations, only asking that the pseudonatural transformation $\pi : \mathcal{E}(-, \mathbf{U}) \rightarrow \mathbb{F}$ induced by the (bicategorical) yoneda

lemma, is a surjection. Namely, this amounts to ask that for every fibration $p : A \rightarrow B \in \mathbb{F}(B)$ over B , there exists a non necessarily unique code $[p] : B \rightarrow \mathbb{U}$ and a map $q : A \rightarrow \tilde{\mathbb{U}}$ making the square a pullback:

$$\begin{array}{ccc} A & \xrightarrow{q} & \tilde{\mathbb{U}} \\ p \downarrow & \lrcorner & \downarrow \pi \\ B & \xrightarrow{[p]} & \mathbb{U} \end{array}$$

In fact, we will require a slightly stronger property for our universe called *realignment*. This property will play a crucial role in showing that our universe hierarchy models cumulativity.

Another thing that we hide under the rug for now is *size issues*. In fact, one must restrict to a suitable class of small fibration if we ought $\mathcal{E}(-, \mathbb{U})$ to be a set and not a proper class. Those issues will be ignored for now, but will be detailed in section 1.3.

First, we still need to generalise our classifier and allow $\pi : \tilde{\mathbb{U}} \rightarrow \mathbb{U}$ to classify a more general notion of *fibred structure*. In some sense, this generalises our core of self-indexing functor.

1.2 Fibred strucutres

We will work in the 2-category $\mathcal{PSH}(\mathcal{E})$ of contravariant pseudo-functors valued in large groupoids $\mathcal{GPD}$ and pseudonatural transformations between them.

Definition 1.2 (Strict Fibration). A strict fibration in $\mathcal{PSH}(\mathcal{E})$ is a *strictly* natural transformation $\mathbb{X} \rightarrow \mathbb{Y}$ such that each component $\mathbb{X}(A) \rightarrow \mathbb{Y}(A)$ is a fibration of groupoids. Similarly, we define strict *discrete* fibrations.

Definition 1.3. A *fibred strucure* on $\mathcal{E}$ is a strict discrete fibration $\phi : \mathbb{F} \rightarrow \mathbb{E}$ (with $\mathbb{E}$ the core of self-indexing functor) that has small fibers. Unfolding this definition, this means that for any map $f : X \rightarrow Y$ in $\mathcal{E}$ we have a set that we call $\mathbb{F}$ -structures on f . And for any cartesian map $f \rightarrow g$ in $\mathcal{E}$ we have a function from the corresponding $\mathbb{F}$ structures.

Example 1.1. If ϕ is simply an inclusion of a subfunctor $\mathbb{F}$, then $\mathbb{F}$ is simply a pullback stable class of maps in $\mathcal{E}$. We call this a *full* notion of fibered structure.

In particular, the inclusion of the functor $\mathbb{F}$ of fibrations in $\mathbb{E}$ forms such a full fibered structure.

Definition 1.4. Local representability A notion of fibered structure $\mathbb{F}$ is called *locally representable* if the strict discrete fibration $\phi : \mathbb{F} \rightarrow \mathbb{E}$ is representable in $\mathcal{PSH}(\mathcal{E})$, in the sense that for any $Z \in \mathcal{E}$ the following weak pullback is equivalent to a representable $\mathcal{E}(-, X)$.

$$\begin{array}{ccc} \mathbb{P} & \longrightarrow & \mathbb{F} \\ \downarrow & \lrcorner & \downarrow \phi \\ \mathcal{E}(-, Z) & \longrightarrow & \mathbb{E} \end{array}$$

Definition 1.5 (Acyclic fibration). A morphism $\mathbb{X} \rightarrow \mathbb{Y}$ in $\mathcal{PSH}(\mathcal{E})$ is an acyclic fibration if it has the right lifting property for all morphism $\mathcal{E}(-, j) : \mathcal{E}(-, A) \rightarrow \mathcal{E}(-, B)$, where $j : A \rightarrow B$ is a cofibration in $\mathcal{E}$.

$$\begin{array}{ccc} \mathcal{E}(-, A) & \longrightarrow & \mathbb{X} \\ \mathcal{E}(-, j) \downarrow & \nearrow & \downarrow f \\ \mathcal{E}(-, B) & \longrightarrow & \mathbb{Y} \end{array}$$

Definition 1.6 (Universe). A *universe* for $\mathbb{F}$, a notion of fibred structure, is a cofibrant object $U \in \mathcal{E}$ together with an acyclic fibration $\pi : \mathcal{E}(U, -) \rightarrow \mathbb{F}$ in $\mathcal{PSH}(\mathcal{E})$. By the Yoneda lemma, this give rise to map $f : \tilde{U} \rightarrow U$ in $\mathcal{E}$ with a chosen $\mathbb{F}$ -structure.

$$\begin{array}{ccc} \mathcal{E}(-, A) & \longrightarrow & \mathcal{E}(-, U) \\ \mathcal{E}(-, j) \downarrow & \dashrightarrow k & \downarrow \pi \\ \mathcal{E}(-, B) & \longrightarrow & \mathbb{Y} \end{array}$$

If we unfold this bicategorical definition in $\mathcal{PSH}$, we get the following factorisation property, which we call *realignment*.

$$\begin{array}{ccc} D & \xrightarrow{\quad} & \tilde{U} \\ \downarrow q & \searrow \lrcorner & \downarrow \pi \\ & E & \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow{h} & U \\ \swarrow i & \downarrow & \searrow k \\ & B & \end{array}$$

1.3 Tackling size issues

Because of the size reasons mentionned previously, we cannot classify the class of fibrations and we must restrict to a class of small fibrations. For this reason, we will restrict $\mathcal{E}$ to be a locally presentable category, of which we now recall the definition.

Definition 1.7 (λ -compact object). We say that $X \in \mathcal{E}$ is λ -compact if $\mathcal{E}(X, -) : \mathcal{E} \rightarrow \mathbf{Set}$ preserves λ -filtered colimits, for λ a regular cardinal.

Definition 1.8 (Locally presentable category). A category $\mathcal{C}$ is locally presentable if it is locally small, has all colimits and has a set of λ -compact objects which generate $\mathcal{C}$ under λ -filtered colimits, for λ a regular cardinal.

Definition 1.9 (Relatively λ -compact morphisms). A morphism $X \rightarrow Y$ in $\mathcal{E}$ is relatively λ -compact if the pullback $Z \times_Y X$ is λ -compact for any morphism $Z \rightarrow Y$, where Z is λ -compact.

Now we recall the following technical result (cf. [1], proposition 4.18).

Theorem 1.1. *Let $\mathcal{E}$ be a locally presentable and locally cartesian closed category. Then there is a κ such that for any $\lambda \triangleright \kappa$, the relatively κ -compact morphisms are a locally representable full notion of fibred structure.*

2 A universe satisfying realignment

We denote $|\mathbb{F}|$ with the image of the map $\phi : \mathbb{F} \rightarrow \mathbb{E}$ such that it is always a *full* notion of fibred structure.

Definition 2.1 (Relative acyclicity). We say that a notion of fibred structure $\mathbb{F}$ is relatively acyclic if the map $\mathbb{F} \rightarrow |\mathbb{F}|$ is an acyclic fibration. This means that for any pullback:

$$\begin{array}{ccc}
X' & \longrightarrow & X \\
f' \downarrow & \lrcorner & \downarrow f \\
Y' & \xrightarrow{i} & Y
\end{array}$$

where f and f' have chosen $\mathbb{F}$ -structure, but the square is not necessarily an $\mathbb{F}$ -morphism, then there exists a new $\mathbb{F}$ structure on f making the square an $\mathbb{F}$ -morphism.

We recall the following definition of Shulman.

Theorem 2.1 (Type-theoretic model toposes). *A type-theoretic model topos is a model category $\mathcal{E}$ which is also :*

- *A Grothendieck 1-topos.*
- *The model structure is right proper, simplicial and combinatorial, and its cofibrations are the monomorphism (right proper simplicial Cisinski model category).*
- *A simplicially locally cartesian closed category.*
- *There is a locally representable and relatively acyclic notion of fibred structure $\mathbb{F}$ such that $|\mathbb{F}|$ is the class of all fibrations.*

Definition 2.2. A realignment data is a pseudo commutative square :

$$\begin{array}{ccccc}
\mathcal{E}(-, A) & \longrightarrow & \mathcal{E}(-, U) & & \\
\downarrow & & \Downarrow \cong & & \downarrow \pi \\
\mathcal{E}(-, A) & \longrightarrow & \mathbb{F} & &
\end{array}$$

We write $D_\kappa(f)$ for the set of chosen representatives of isomorphism classes of realignment data for f . This set is μ -small as long as U is μ -compact.^a

^are-check that

Lemma 2.2. *Any locally representable notion of fibred structure $\phi : \mathbb{F} \rightarrow \mathbb{E}$ preserves coproducts, pushout of cofibrations and transfinite composite of cofibrations if $\mathbb{E}$ does.*

Lemma 2.3. *Let's assume that $\mathcal{E}$ is cofibrantly generated by $\mathcal{I}$ and $\mathbb{X}, \mathbb{Y}$ preserves coproducts, pushout of cofibrations and transfinite composite of cofibrations. If $f : \mathbb{X} \rightarrow \mathbb{Y}$ satisfies the realignment property with respect to the cofibrations in $\mathcal{I}$, then f is an acyclic fibration.*

Theorem 2.4 (Universe construction). *Let $\mathbb{F}$ be a locally representable and relatively acyclic notion of fibred structure on $\mathcal{E}$, then $|\mathbb{F}|$ has a universe.*

Proof. We want to find an acyclic fibration $f : \mathcal{E}(-, U) \rightarrow \mathbb{F}$ with $U \in \mathcal{E}$ a cofibrant object, which will be our universe. To construct it, we use a bicategorical version of the small object argument. As we are in a type-theoretic model topos, our cofibrations are monomorphisms, of which we have a λ -small generating set $\mathcal{I}$ whose domains and codomains are λ -compact for any strongly inaccessible cardinal $\lambda > \kappa$ for the κ defined in 1.1.

We will define a cell-complex structure indexed on the regular ordinals $\mathcal{O}$:

$$U_\lambda^0 \longrightarrow \dots \longrightarrow U_\lambda^\alpha \xrightarrow{j_{\alpha, \alpha+1}} U_\lambda^{\alpha+1} \longrightarrow \dots$$

with maps $f_\lambda^\alpha : \mathcal{E}(-, \mathcal{U}_\lambda^\alpha) \rightarrow \mathbb{F}$ and coherence isomorphisms $f_\lambda^\alpha \circ j_{\beta, \alpha} \cong f_\lambda^\beta$. This structure is defined by transfinite induction on $\alpha \in \mathcal{O}$, we detail all the cases in the following.

Base case. We take $\mathcal{U}_\lambda^0 := \emptyset$ with the evident map $f_\lambda^0 : \mathcal{E}(-, \emptyset) \rightarrow \mathbb{F}$.

Successor case. Suppose $\mathcal{U}_\lambda^\alpha$ and f_λ^α are defined. We construct $\mathcal{U}_\lambda^{\alpha+1}$ as the following bicategorical pushout:

$$\begin{array}{ccc} \coprod_{d \in D(f_\kappa^\alpha)} A_d & \longrightarrow & \mathcal{U}_\kappa^\alpha \\ \downarrow & \Downarrow \cong & \downarrow \\ \coprod_{d \in D(f_\kappa^\alpha)} B_d & \dashrightarrow & \mathcal{U}_\kappa^{\alpha+1} \end{array}$$

As $\mathbb{F}$ preserves coproducts and pushouts of monomorphisms because it is a stack for cell-complexes (lemma 2.2), we get an essentially unique induced map $f_\lambda^{\alpha+1} : \mathcal{E}(-, \mathcal{U}_\lambda^{\alpha+1}) \rightarrow \mathbb{F}$, along with the corresponding coherence isomorphisms.

Limit case. If f_λ^β is defined for all $\beta < \alpha$, with α a limit ordinal, then we take $\mathcal{U}_\lambda^\alpha := \operatorname{colim}_{\beta < \alpha} \mathcal{U}_\lambda^\beta$. Again, $\mathbb{F}$ preserves transfinite composition of monomorphisms (lemma 2.2) and thus there is an essentially unique induced map $f_\lambda^\alpha : \mathcal{E}(-, \mathcal{U}_\lambda^\alpha) \rightarrow \mathbb{F}$, along with the corresponding coherence isomorphisms.

We can stop the transfinite iteration at κ , and define $\mathcal{U}_\lambda := \operatorname{colim}_{\alpha < \kappa} \mathcal{U}_\lambda^\alpha$ along with the corresponding $\pi_\lambda : \mathcal{E}(-, \mathcal{U}_\lambda) \rightarrow \mathbb{F}$ which is the essentially unique map as above in the construction. The latter satisfies the right lifting property against $\mathcal{I}$, and is thus an acyclic fibration by lemma 2.3. This means that we get generic maps $\pi_\lambda : \widetilde{\mathcal{U}}_\lambda \rightarrow \mathcal{U}_\lambda$ by the bicategorical Yoneda lemma, which are universes in the sense that they satisfy the realignment property (definition 1.6). $\square$

We summarize everything in the following theorem, and prove that the constructed universes are fibrant and univalent.

Theorem 2.5 (Univalent Universes). *Let $\mathcal{E}$ be a type theoretic model topos. There is a regular cardinal κ such that for any regular cardinal $\lambda \triangleright \kappa$, there exists a morphism $\pi_\lambda : \widetilde{\mathcal{U}}_\lambda \rightarrow \mathcal{U}_\lambda$ such that:*

- *The λ -presentable objects in $\mathcal{E}$ are closed under finite limits.*
- *$\pi_\lambda : \widetilde{\mathcal{U}}_\lambda \rightarrow \mathcal{U}_\lambda$ is a relatively λ -compact morphism equipped with an $|\mathbb{F}|$ structure.*
- *Every relatively λ -compact morphism with a $|\mathbb{F}|$ structure is a pullback of π_λ .*
- *$\mathcal{U}_\lambda$ is a fibrant object.*
- *π_λ satisfy the univalence axiom.*

Proof. Of course, κ is given by theorem 1.1 and the map π_λ is given by the previous lemma.¹ $\square$

3 Universe inclusions and cumulativity

Definition 3.1. A diagram $J : \mathcal{D} \rightarrow \mathcal{E}$ is said to satisfy descent when for any cartesian natural transformation $\alpha : K \rightarrow J$, the induced morphisms $\alpha_d \rightarrow \operatorname{colim}_{\mathcal{D}} \alpha_d$ in $\mathcal{E}^{\rightarrow}$ are cartesian for each $d \in \mathcal{D}$.

¹Careful: applied to what ?

$$\begin{array}{ccc}
J(d) & \longrightarrow & \operatorname{colim}_{\mathcal{D}} J \\
\alpha_d \downarrow & \lrcorner & \downarrow \operatorname{colim}_{d \in \mathcal{D}} \alpha_d \\
K(d) & \longrightarrow & \operatorname{colim}_{\mathcal{D}} K
\end{array}$$

Lemma 3.1. *Let $F, G : \mathcal{D} \rightarrow \mathcal{E}$ be two diagrams such that G satisfies descent, and let $F \rightarrowtail G$ be a cartesian monomorphism. Then the induced map $\operatorname{colim}_{\mathcal{D}} F \rightarrow \operatorname{colim}_{\mathcal{D}} G$ is itself a monomorphism.*

Theorem 3.2 (Injective Lifts). *There exists a cartesian monomorphism $\pi_\lambda \rightarrowtail \pi_\mu$ for $\kappa < \lambda < \mu$ two strongly inaccessible cardinals.*

Proof. We are going to construct a cartesian monomorphism $U_\lambda \xrightarrow{\uparrow_\lambda^\mu} U_\mu$ between the universe objects for $\kappa < \lambda < \mu$. First, recall that each U_λ is a *filtered* colimit : $U_\lambda = \operatorname{colim}_{\alpha < \kappa} U_\lambda^\alpha$. As filtered colimits enjoy descent,

to construct the monomorphism $\operatorname{colim}_{\alpha < \kappa} U_\lambda^\alpha \xrightarrow{\uparrow_\lambda^\mu} \operatorname{colim}_{\alpha < \kappa} U_\mu^\alpha$, by lemma 3.1 it suffices to construct a monomorphism of (bicategorical) diagrams $U_\lambda^\bullet \rightarrowtail U_\mu^\bullet$. Therefore, we need to deal with two cases.

Successor case We need to define $U_\lambda^{\alpha+1} \rightarrowtail U_\mu^{\alpha+1}$ given $U_\lambda^\alpha \rightarrowtail U_\mu^\alpha$. According to the previous construction, $U_\bullet^{\alpha+1}$ is constructed using pushouts and coproducts along cofibrations, it is the colimit of a diagram that satisfies descent. Because of lemma 3.1, it suffices to construct a monomorphism between the defining realignment data of $U_\lambda^{\alpha+1}$ and $U_\mu^{\alpha+1}$. Such a morphism is induced by the embedding that send a realignment data for U_λ^α to a realignment data for $U_\lambda^{\alpha+1}$ by the following composition.

$$\begin{array}{ccccc}
\mathcal{E}(-, A) & \longrightarrow & \mathcal{E}(-, U_\lambda^\alpha) & \xrightarrow{\mathcal{E}(-, U_\lambda^\alpha \rightarrow U_\lambda^{\alpha+1})} & \mathcal{E}(-, U_\lambda^{\alpha+1}) \\
\downarrow & & \Downarrow \cong & \downarrow \pi_\lambda^\alpha & \downarrow \cong \\
\mathcal{E}(-, B) & \longrightarrow & \mathbb{F} & \xleftarrow{\pi_\lambda^{\alpha+1}} &
\end{array}$$

Where the right triangle pseudo-commutes because of the coherence isomorphisms of the small object construction (theorem 2.4).

Limit case. If $U_\lambda^\beta \rightarrowtail U_\mu^\beta$ is defined for all $\beta < \alpha$, then again by lemma 3.1 we have a monomorphism $\operatorname{colim}_{\beta < \alpha} U_\lambda^\beta \rightarrowtail \operatorname{colim}_{\beta < \alpha} U_\mu^\beta$ i.e. $U_\lambda^\alpha \rightarrowtail U_\mu^\alpha$. $\square$

4 Coherence from local universes

TODO : present the local universes construction, and choice of codes through realignment.

References

- [1] Michael Shulman. “All $(\infty, 1)$ -toposes have strict univalent universes”. In: (2019). URL: <https://arxiv.org/abs/1904.07004>.